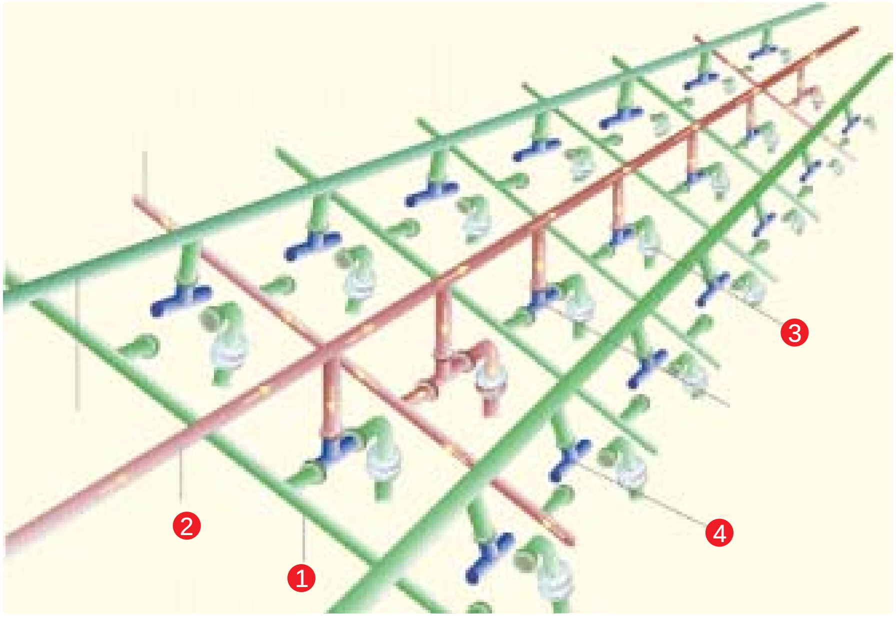

4 **Closed or open transistor:** It holds the data signal which can either be 1 or 0 when the transistor is closed. If it is open, it is disconnected from the series, and hence the system bus, and it could be in either state. This is called floating output.

Do Remember

- Enable the lowest CAS setting for your RAM in the BIOS. Also, set it to use the highest bus speed setting it supports.
- Close any applications you do not need so as to make more memory available for more important programs.

Buying Tips

✓ **MINIMUM RAM:** Most modern applications as well as operating systems such as Windows XP need at least 256 MB of RAM to run smoothly.

✓ **DDR OR SD RAM:** If your motherboard supports both DDR-RAM and SDRAM, opt for the former. Though the prices are nearly the same, the performance of DDR-RAM is far superior.

✓ **FREQUENCY:** The frequency of the RAM has to match the one supported by the motherboard.

✓ **SINGLE OR MULTIPLE:** As far as possible, use a single RAM module. The advantages are manifold. Firstly, a single stick costs less than multiple modules put together. Secondly, there won't be any compatibility issues, and you still have a vacant slot for a further upgrade.

✓ If you're buying a new motherboard that's based on the latest chipsets (Nforce2, 865, 875, etc.), opt for 400 MHz memory modules, which are currently the fastest available.

✓ Avoid Investing in SD-RAM. Not only will they be outdated soon, they're also more expensive than the better and faster DDR-RAM. You could also have warranty problems, as most dealers don't offer support for it anymore.

✓ Always buy branded RAM modules sold with proper packaging. Lifetime (5-year) warranties are also offered, albeit at a higher price.

1 **Address Line:** It carries either a 0 or a 1 signal amongst the transistors. There are two address lines for a series of transistors. If one address line is carrying the signal 1, then the other address line carries the signal 0.

2 **Data line:** It carries the cumulative data signal from the RAM to the system bus.

3 **Capacitor:** It holds the information—either 0 or 1—depending on whether it is in a charged or discharged state.

Quick Find

Basic Use	YOU NEED	To upgrade RAM on an older board that has no Support for DDR-RAM
	LOOK FOR	SDRAM that preferably has a frequency of 133 MHz
Inter Use	YOU NEED	A cost-effective solution
	LOOK FOR	DDR-RAM that has a frequency in the range of 266 MHz to 333 MHz
Prof Use	YOU NEED	A performer—cost no bar
	LOOK FOR	400 MHz DDR-RAM

The Future

More often than not, the focus in RAM is mostly on speed, with one manufacturer constantly trying to outdo the other. The DIMM has not seen much change recently. All that has been included in the design is the HeatSpreader to boost the cooling process.

All in all, the DRAM module isn't scheduled for a massive change in the desktop segment.

However, the server segment witnesses an ever-increasing demand for RAM with practically no space for expansion. This has seen the emergence of some exciting new technologies such as Thin Small Outline Packages (TSOP), Tape Carrier Packaging (TCP), Elevated Package Over CSP (EPOC) and Foldable Electronic Memory Module Assembly (FEMMA) to overcome the problem.

The evolution of RAM modules, from the slow EDO-RAMs to the RD-RAM, and now DDR-RAM, has always strived to push the speed limit a little further. This will continue to be a trend in the future as well. While DDR 400 MHz is still the most widely accepted and recognised standard, memory manufacturers have succeeded in churning out faster memory, running at 533 MHz. With the advent of 64-bit processors that require ECC memory, the market will witness another major change. These ECC memory modules, which were only used in servers thus far, will trickle down to the desktop arena.